

Consumption of artificial sweetener– and sugar-containing soda and risk of lymphoma and leukemia in men and women

Background: Despite safety reports of the artificial sweetener aspartame, health-related concerns remain. Objective: We prospectively evaluated whether the consumption of aspartame- and sugar-containing soda is associated with risk of hematopoietic cancers. Design: We repeatedly assessed diet in the Nurses' Health Study (NHS) and Health Professionals Follow-Up Study (HPFS). Over 22 y, we identified 1324 non-Hodgkin lymphomas (NHLs), 285 multiple myelomas, and 339 leukemias. We calculated incidence RRs and 95% CIs by using Cox proportional hazards models. Results: When the 2 cohorts were combined, there was no significant association between soda intake and risks of NHL and multiple myeloma. However, in men, ≥ 1 daily serving of diet soda increased risks of NHL (RR: 1.31; 95% CI: 1.01, 1.72) and multiple myeloma (RR: 2.02; 95% CI: 1.20, 3.40) in comparison with men who did not consume diet soda. We observed no increased risks of NHL and multiple myeloma in women. We also observed an unexpected elevated risk of NHL (RR: 1.66; 95% CI: 1.10, 2.51) with a higher consumption of regular, sugar-sweetened soda in men but not in women. In contrast, when sexes were analyzed separately with limited power, neither regular nor diet soda increased risk of leukemia but were associated with increased leukemia risk when data for men and women were combined (RR for consumption of ≥ 1 serving of diet soda/d when the 2 cohorts were pooled: 1.42; 95% CI: 1.00, 2.02). Conclusion: Although our findings preserve the possibility of a detrimental effect of a constituent of diet soda, such as aspartame, on select cancers, the inconsistent sex effects and occurrence of an apparent cancer risk in individuals who consume regular soda do not permit the ruling out of chance as an explanation.